Jeonghyeon Kim

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Research Interest

My primary research goal is to build reliable and trustworthy AI systems without compromising their predictive performance. To achieve this, I actively conduct research on Out-of-Distribution (OoD) detection, machine unlearning, multi-modal fine-tuning, and concept erasure. I particularly enjoy interpreting these problems through the lens of Energy-Based Models (EBMs) to provide theoretical grounding for empirical results.

Education

Seoul National University of Science and Technology (SeoulTech) Seoul, South Korea
Ph.D. Student in Data Science (Advisor: Prof. Sangheum Hwang) *Mar. 2023 - present*

Seoul National University of Science and Technology (SeoulTech) Seoul, South Korea
B.S. in Industrial Engineering *Mar. 2016 - Feb. 2022*

Research Experience

DAINTLAB, SeoulTech Seoul, South Korea
Research Assistant (Advisor: Prof. Sangheum Hwang) *Aug. 2022 - present*

- **Generative AI Safety & Concept Erasure**:
  - Addressed the trade-off between removing harmful concepts and preserving generation quality in Diffusion Models. Proposed **High-level Representation Misdirection (HiRM)**, a method for localized concept erasure by misdirecting semantic representations in the text encoder.
  - Developed a strategy to update only early layers while guiding high-level semantics, achieving precise removal of target concepts (e.g., styles, objects, NSFW) with minimal impact on unrelated concepts.
  - ***Selected Publication: Accepted at ICLR 2026***

- **Trustworthy Multimodal Representation Learning**:
  - Proposed **Cross-Modal Alignment (CMA)** to enhance Out-of-Distribution (OoD) detection in Vision-Language Models (VLMs) like CLIP.
  - Theoretically demonstrated that the CMA regularization corresponds to maximizing the likelihood of a joint Energy-Based Model (EBM) on a hyperspherical space.
  - Achieved state-of-the-art OoD detection performance by mitigating the modality gap within ID embeddings while maintaining high ID accuracy.
  - ***Selected Publication: Accepted at CVPR 2025***

Machine Learning Data Research Group, KISTI Daejeon, South Korea
Research Intern (Advisor: Dr. Ryong Lee) *Jul. 2021 - Aug. 2021*

- **Privacy-Preserving Pipeline for Autonomous Driving**:
  - Built a data de-identification pipeline using object detection (YOLOv3) to protect personal information (faces, license plates) in large-scale driving datasets.

Skills

Languages & Frameworks	Linux, Python, PyTorch, Hugging Face, Docker
Research Areas	Concept Erasure, Machine Unlearning, OoD Detection, Multimodal Learning

Publications

International Conferences

[**IC4**] *"Localized Concept Erasure in Text-to-Image Diffusion Models via High-Level Representation Misdirection"*
Uichan Lee*, **Jeonghyeon Kim*** and Sangheum Hwang
International Conference on Learning Representations (**ICLR**), 2026

[**IC3**] *"Enhanced OoD Detection through Cross-Modal Alignment of Multi-Modal Representations"*
Jeonghyeon Kim and Sangheum Hwang
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025

[**IC2**] *"Comparison of Out-of-Distribution detection Performance of CLIP-based Fine-Tuning Methods"*
Jeonghyeon Kim, Jihyo Kim, and Sangheum Hwang
International Conference on Electronics, Information, and Communication (**ICEIC**), 2024 (**Oral, Best Paper Award**)

[**IC1**] *"Deep Active Learning with Contrastive Learning Under Realistic Data Pool Assumptions"*
Jihyo Kim, **Jeonghyeon Kim**, and Sangheum Hwang
AAAI Workshop on Practical Deep Learning in the Wild (**AAAI-W**), 2023

Domestic Conferences

[**DC5**] *"Reference-free LLM Unlearning Evaluation through Energy Trajectory Analysis"*
Jeonghyeon Kim, and Sangheum Hwang
Korea Computer Congress (**KCC**), 2026 (**Oral, Best Paper Award**)

[**DC4**] *"Parameter-Efficient LLM Unlearning through Sparse Attention Head Updates"*
Deokyong Lee, **Jeonghyeon Kim**, and Sangheum Hwang
Korean Institute of Industrial Engineers (**KIIE**), 2025

[**DC3**] *"Uncovering Hidden Vulnerabilities in Machine Unlearning: Adversarial Attack as a Probe and Pruning as a Solution"*
Hwiyeong Lee, **Jeonghyeon Kim**, and Sangheum Hwang
Korea Computer Congress (**KCC**), 2024

[**DC2**] *"Uniformity Reduction Regularization: Multimodal Fine-tuning to Enhance Out-of-Distribution Detection Performance"*
Jeonghyeon Kim and Sangheum Hwang
Korea Computer Congress (**KCC**), 2024 (**Oral**)

[**DC1**] *"Enhancing Out-of-Distribution Detection Performance of CLIP Based on Fine-tuning with Random Texts"*
Jeonghyeon Kim, Jihyo Kim, and Sangheum Hwang
Korea Computer Congress (**KCC**), 2023 (**Oral, Best Paper Award**)

Invited Talks

KCC 2026	Enhanced OoD Detection through Cross-Modal Alignment *(Jeju, ICC Jeju)*	*Jun. 2026*
KCCV 2025	Enhanced OoD Detection through Cross-Modal Alignment *(Busan, Bexco)*	*Jul. 2025*
KSIIM 2025	Enhanced OoD Detection through Cross-Modal Alignment *(Seoul, Seoul National University College of Medicine)*	*Feb. 2025*

Honors and Awards

Best Paper Award	KCC 2026	*Jun. 2026*
Ph.D. Student Research Grant	National Research Foundation of Korea (NRF)	*Aug. 2025*

- Grant Amount: 50M KRW / 2 years

- Topic: Mechanistic Interpretability of Knowledge Unlearning in Foundation Models

Best Paper Award	ICEIC 2024	*Jan. 2024*
Best Paper Award	KCC 2023	*Jun. 2023*
Gold Prize	Capstone Competition, Dept. of IISE, SeoulTech	*Dec. 2021*
School Honor Scholarship	SeoulTech	*Fall 2021*
Academic Excellence Scholarship	SeoulTech (4 semesters)	*2016 - 2021*

Licenses and Certifications

Engineer Quality Management	Human Resources Development Service of Korea	*Dec. 2020*
Engineer Information Processing	Human Resources Development Service of Korea	*May. 2019*

Teaching Experience

Neural Networks and Deep Learning	SeoulTech
Teaching Assistant (Instructor: Prof. Sangheum Hwang)	*Spring 2026*
Deep Learning	SeoulTech
Teaching Assistant (Instructor: Prof. Sangheum Hwang)	*Spring 2024*
Artificial Intelligence	SeoulTech
Teaching Assistant (Instructor: Prof. Sangheum Hwang)	*Fall 2023*